

Named Entity Recognition

The named entity recognition (NER) is one of the most popular data preprocessing task. It involves the identification of key information in the text and classification into a set of predefined categories. An entity is basically the thing that is consistently talked about or refer to in the text.

NER is the form of NLP.

At its core, NLP is just a two-step process, below are the two steps that are involved:

1. Detecting the entities from the text
2. Classifying them into different categories

Some of the categories that are the most important architecture in NER such that:

1.Person 2.Organization 3.Place/ location Other common tasks include classifying of the following: List item

1. date/time.
2. Numeral measurement (money, percent, weight, etc)
3. E-mail address

Implementation

In this implementation, we will perform Named Entity Recognition using two different frameworks: Spacy and NLTK. This code can be run on colab, however for visualization purpose. I recommend the local environment. We can install the following frameworks using pip install First, we performed Named Entity recognition using Spacy.

```
# command to run before code
! pip install spacy
! pip install nltk
! python -m spacy download en_core_web_sm

# imports and load spacy english language package
import spacy
from spacy import displacy
from spacy import tokenizer
nlp = spacy.load('en_core_web_sm')
```

```
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/pyt
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/pyt
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/pytho
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/
Requirement already satisfied: wasabi<2.1.0,>=2.0.0 in /usr/local/lib/python3.12/
```

```

Requirement already satisfied: presnet<3.1.0,>=3.0.2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: weasel<0.5.0,>=0.1.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: typer<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pydantic!=1.8,!<1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: langcodes<4.0.0,>=3.2.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: language-data>=1.2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pydantic-core==2.33.2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: typing-extensions>=4.12.2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: typing-inspection>=0.4.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: click>=8.0.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: rich>=10.11.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: marisa-trie>=1.1.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
Collecting en-core-web-sm==3.8.0

```

Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0.tar.gz 12.8/12.8 MB 94.1 MB/s eta 0:00:00

✓ Download and installation successful

You can now load the package via `spacy.load('en_core_web_sm')`

⚠ Restart to reload dependencies

If you are in a Jupyter or Colab notebook, you may need to restart Python in order to load all the package's dependencies. You can do this by selecting the 'Restart kernel' or 'Restart runtime' option.

```

import spacy
from spacy import displacy
from spacy import tokenizer

```

```
from spacy import tokenize
nlp = spacy.load('en_core_web_sm')

#Load the text and process it
# I copied the text from python wiki

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# I copied the text from python wiki
text = "World War II[b] or the Second World War (1 September 1939 - 2 September 1945) was a global conflict between two coalitions: the Allies and the Axis powers."
text2 = # copy the paragraphs from https://www.python.org/doc/essays/
doc = nlp(text)
#doc2 = nlp(text2)
sentences = list(doc.sents)
print(sentences)
# tokenization
for token in doc:
    print(token.text)
# print entities
ents = [(e.text, e.start_char, e.end_char, e.label_) for e in doc.ents]
print(ents)
# now we use displacy function on doc2
displacy.render(doc, style='ent', jupyter=True)
# text2 = # copy the paragraphs from https://www.python.org/doc/essays/
```

```
[World War II[b] or the Second World War (1 September 1939 - 2 September 1945) was a global conflict between two coalitions: the Allies and the Axis powers.]
World
War
II[b
]
or
the
Second
World
War
(
1
September
1939
-
2
September
1945
)
was
a
global
conflict
between
two
coalitions
:
the
```

Allies
and
the
Axis
powers
.
Nearly
all
of
the
world
's
countries
participated
,
with
many
nations
mobilising
all
resources
in
pursuit
of
total
war
.
Tanks
and
aircraft
played
major
roles
,
enabling
the
strategic
bombing
of
cities
and
delivery
of
the
first
and
only
nuclear
weapons
ever
used
in
war
.
..

world
war
II
is
the
deadliest
conflict
in
history
,
causing
the
death
of
70
to
85
million
people
,
more
than
half
of
whom
were
civilians
. Millions
died
in
genocides
,
including
the
Holocaust
,
and
by
massacres
,
starvation
,
and
disease
. After
the
Allied
victory
,
Germany
,
Austria

,
Japan
,
and
Korea
were
occupied
,
and
German
and
Japanese
leaders
were
tried
for
war
crimes
.
The
causes
of
World
War
II
included
unresolved
tensions
in
the
aftermath
of
World
War
I
,
the
rise
of
fascism
in
Europe
and
militarism
in
Japan
.
Key
events
preceding
the
war
included
Japan
's

invasion
of
Manchuria
in
1931
,
the
Spanish
Civil
War
,
the
outbreak
of
the
Second
Sino
-
Japanese
War
in
1937
,
and
Germany
's
annexations
of
Austria
and
the
Sudetenland
.
World
War
II
is
generally
considered
to
have
begun
on
1
September
1939
,
when
Nazi
Germany
,
under
Adolf
Hitler

,
invaded
Poland

,
after
which
the
the
United
Kingdom
and
France
declared
war
on
Germany

. Poland
was
divided
between
Germany
and
the
Soviet
Union
under
the
Molotov
–
Ribbentrop
Pact

. In
1940

,
the
Soviet
Union
annexed
the
Baltic
states
and
parts
of
Finland
and
Romania

. After
the
fall
of
France
.

in
June
1940
,
the
war
continued
mainly
between
Germany
and
the
British
Empire
,
with
fighting
in
the
Balkans
,
Mediterranean
,
and
Middle
East
,
the
aerial
Battle
of
Britain
and
the
Blitz
,
and
the
naval
Battle
of
the
Atlantic
.
Through
campaigns
and
treaties
,
Germany
gained
control
of
much
of

,
continental
Europe
and
formed
the
Axis
alliance
with
Italy
,
Japan
,
and
other
countries

.
In
June
1941

,
Germany
invaded
the
Soviet
Union

,
opening
the
Eastern
Front
and
initially
making
large
territorial
gains

.
[('World War II[b', 0, 14, 'EVENT'), ('the Second World War', 19, 39, 'EVENT'),
World War II[b EVENT] or the Second World War EVENT (1 September 1939 DATE –
2 September 1945 DATE) was a global conflict between two coalitions: the Allies and the Axis
powers. Nearly all of the world's countries participated, with many nations mobilising all resources
in pursuit of total war. Tanks and aircraft played major roles, enabling the strategic bombing of
cities and delivery of the first ORDINAL and only nuclear weapons ever used in war. World
War II EVENT is the deadliest conflict in history, causing the death of 70 to 85 million MONEY
people, more than half CARDINAL of whom were civilians. Millions CARDINAL died in
genocides, including the Holocaust EVENT , and by massacres, starvation, and disease. After

the Allied **ORG** victory, Germany **GPE** , Austria **GPE** , Japan **GPE** , and Korea **GPE** were occupied, and German **NORP** and Japanese **NORP** leaders were tried for war

1. Explain NER

It is a task that identifies and classifies key information in text into categories like Person, Organization, Location, Date/Time, etc. It improves search by understanding queries (Relationship between words)

2. Application of NLP

- Translation
- Transcription
- Sentiment Analysis
- Speech Recognition
- Chatbots

3. Short note on Tokenization

Tokenization is the process of converting Text/paragraphs into smaller units called as tokens. The first step is to preprocess the text - remove the stopwords, perform stemming and lemmatization, and then we get the tokens.

